

Query Processing And Optimization In Distributed Databases

Suppose that the EMPLOYEE and DEPARTMENT relations in Figure 3.5 are distributed at two sites as shown in Figure 25.10. We will assume in this example that neither relation is fragmented. According to Figure 25.10, the size of the EMPLOYEE relation is $100 * 10,000 = 106$ bytes, and the size of the DEPARTMENT relation is $35 * 100 = 3500$ bytes.

Figure 25.10

Example to illustrate volume of data transferred.

Site 1:

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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10,000 records

each record is 100 bytes long

Ssn field is 9 bytes long

Dno field is 4 bytes long

Fname field is 15 bytes long

Lname field is 15 bytes long

Site 2:

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
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100 records

each record is 35 bytes long

Dnumber field is 4 bytes long

Mgr_ssn field is 9 bytes long

Dname field is 10 bytes long

Consider the query Q: For each employee, retrieve the employee name and the name of the department for which the employee works. This can be stated as follows in the relational algebra:

$$Q: \pi_{Fname, Lname, Dname} (EMPLOYEE \bowtie_{Dno=Dnumber} DEPARTMENT)$$

The query is submitted at a distinct site 3, which is called the result site because the query result is needed there. Neither the EMPLOYEE nor the DEPARTMENT relations reside at site 3. There are three simple strategies for executing this distributed query:

Transfer both the EMPLOYEE and the DEPARTMENT relations to the result site, and perform the join at site 3. In this case, a total of $1,000,000 + 3,500 = 1,003,500$ bytes must be transferred.

Transfer the EMPLOYEE relation to site 2, execute the join at site 2, and send the result to site 3. The size of the query result is $40 * 10,000 = 400,000$ bytes, so $400,000 + 1,000,000 = 1,400,000$ bytes must be transferred.

Transfer the DEPARTMENT relation to site 1, execute the join at site 1, and send the result to site 3. In this case, $400,000 + 3,500 = 403,500$ bytes must be transferred.

If minimizing the amount of data transfer is our optimization criterion, we should choose strategy. Now consider another query Q: For each department, retrieve the department name and the name of the department manager. This can be stated as follows in the relational algebra:

$$Q': \pi_{Fname, Lname, Dname} (DEPARTMENT \bowtie_{Mgr_ssn=Ssn} EMPLOYEE)$$

Again, suppose that the query is submitted at site 3. The same three strategies for executing query Q apply to Q, except that the result of Q includes only 100 records, assuming that each department has a manager:

.Transfer both the EMPLOYEE and the DEPARTMENT relations to the result site, and perform the join at site 3. In this case, a total of $1,000,000 + 3,500 = 1,003,500$ bytes must be transferred.

Transfer the EMPLOYEE relation to site 2, execute the join at site 2, and send the result to site 3. The size of the query result is $40 * 100 = 4,000$ bytes, so $4,000 + 1,000,000 = 1,004,000$ bytes must be transferred.

Transfer the DEPARTMENT relation to site 1, execute the join at site 1, and send the result to site 3. In this case, $4,000 + 3,500 = 7,500$ bytes must be transferred.

Again, we would choose strategy 3 this time by an overwhelming margin over strategies 1 and 2.

The preceding three strategies are the most obvious ones for the case where the result site (site 3) is different from all the sites that contain files involved in the query (sites 1 and 2). However, suppose that the result site is site 2; then we have two simple strategies: Transfer the EMPLOYEE relation to site 2, execute the query, and present the result to the user at site 2. Here, the same number of bytes 1,000,000 must be transferred for both Q and Q'.

Transfer the DEPARTMENT relation to site 1, execute the query at site 1, and send the result back to site 2. In this case $400,000 + 3,500 = 403,500$ bytes must be transferred for Q and $4,000 + 3,500 = 7,500$ bytes for Q'.